Hakai
Science on the Coastal Margin

Metadata Report - Time series of eelgrass meadow extent derived from remotely piloted aerial system (RPAS, or drone) surveys, Central Coast, British Columbia

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Last edited: 2026-06-11

Abstract

This data package represents a time series of eelgrass (*Zostera marina*) meadow extent derived from remotely piloted aerial system (RPAS or drone) surveys, along with relevant metadata. RPAS surveys are conducted annually at monitoring sites on the Central Coast of British Columbia, Canada. These data are collected as part of the Hakai Institute Habitat Mapping Program, whose broader goal is to document and understand long-term trends in eelgrass meadow dynamics and drivers at local, regional, and coast-wide scales. The Hakai Institute started using drones in 2015 as part of this work in order to capture site-level data on eelgrass distribution for long-term ecological research.

Drone surveys are conducted annually in July/August during low tide (<0.5 m, chart datum). Orthomosaics are generated using a Structure from Motion Multi-View Stereo (SfM-MVS) workflow (Pix4Dmapper, DroneDeploy, or Agisoft Metashape Pro) and georeferenced using ground control points collected during the survey or co-registered to a reference orthomosaic. Eelgrass extent is delineated using a combination of object-based image analysis (OBIA) in eCognition Developer 9 and manual delineation in ArcMap (v10.8). Segmentation outputs are imported into ArcGIS, where objects are classified as eelgrass by a trained analyst based on photo-interpretive characteristics. Where segmentation performs poorly, segments are removed and redrawn manually. Where available, towed underwater video data are used to aid image interpretation. A minimum mapping unit (MMU) of 2 m^2 is applied. All patches within a 1 m distance of their neighbour are aggregated.

Areal extent data are provided as vector polygon features in NAD83 UTM Zone 9N, clipped to each site area of interest (AOI) to ensure consistent survey boundaries are compared over time, and published to a geodatabase. For further details on factors affecting mapping confidence, see Nahirnick et al. (2018).

This data package includes:

1. A geodatabase and geopackage which include:

    - Eelgrass extent polygons — annual delineations of *Z. marina* meadow extent at each monitoring site (Eelgrass_Drone_extents_AOIClipped)
    - Area of interest (AOI) polygons — consistent survey boundaries enabling temporal comparison (Eelgrass_Drone_AOIs)

2. This metadata report (.pdf) describes methods for imagery collection, generating orthomosaics and delineating eelgrass extent.

3. Data dictionaries (.csv) which describe the attributes of the polygon vector features.

GENERAL INFORMATION

Title of Dataset: Time Series of Eelgrass (*Zostera marina*) Extent Derived from Drone Surveys, Central Coast, British Columbia

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Alternate Contact(s): data@hakai.org

Date of data collection: 2015-07-15 - present

Geographic location of data collection: Central Coast, British Columbia, Canada

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SHARING/ACCESS INFORMATION

This data package is freely available to everyone, following the principles of equitable access and benefit sharing. However, we expect all data users to give attribution to the data providers (read our data license) and the use of these data should happen in the light of fair use, i.e.: 1) respect the data providers, and provide helpful feedback on data quality, and 2) communicate and/or collaborate with the providers if you are considering using this dataset for manuscripts or other forms of reporting.

Recommended citation: Reshitnyk, L., Guyn, A., Holmes, K., & Mai, T. (2026). *Eelgrass (Z. marina) extent at sites along the Central Coast, British Columbia* (current version). Hakai Institute. https://doi.org/10.21966/03pw-2190. Date accessed:

DATA & FILE OVERVIEW
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1. Layer Name: Hakai_Drone_AOIs
   a. Feature Class: Eelgrass_Drone_AOIs
   b. Type: Polygon
   c. Number of Features: 8
   d. Fields (5):
      - OBJECTID (OID)
      - Shape (Geometry)
      - Shape_Area (Double)
      - Shape_Length (Double)
      - siteName (String)
2. Layer Name: Eelgrass_Drone_extents_AOIClipped
   a. Feature Class:
   b. Type: Polygon
   c. Number of Features: 55
   d. Fields (25):
      - hab_filena (String)
      - skp_site_n (String)
      - skp_img_da (Date)
      - skp_repeat (String)
      - skp_region (String)
      - skp_data_p (String)
      - skp_img_sr (String)
      - skp_sensor (String)
      - skp_mobe (String)
      - skp_tide_m (String)
      - skp_img_re (String)
      - skp_descri (String)
      - skp_habtyp (String)
      - skp_sharin (String)
      - skp_contac (String)
      - skp_DOI (String)
      - skp_specie (String)
      - skp_delin_ (String)
      - skp_conf (String)
      - skp_analys (String)
      - skp_notes (String)
      - skp_Last_M (String)
      - skp_Last_1 (String)
      - Shape_Leng (Float)

- Shape_Area (Float)

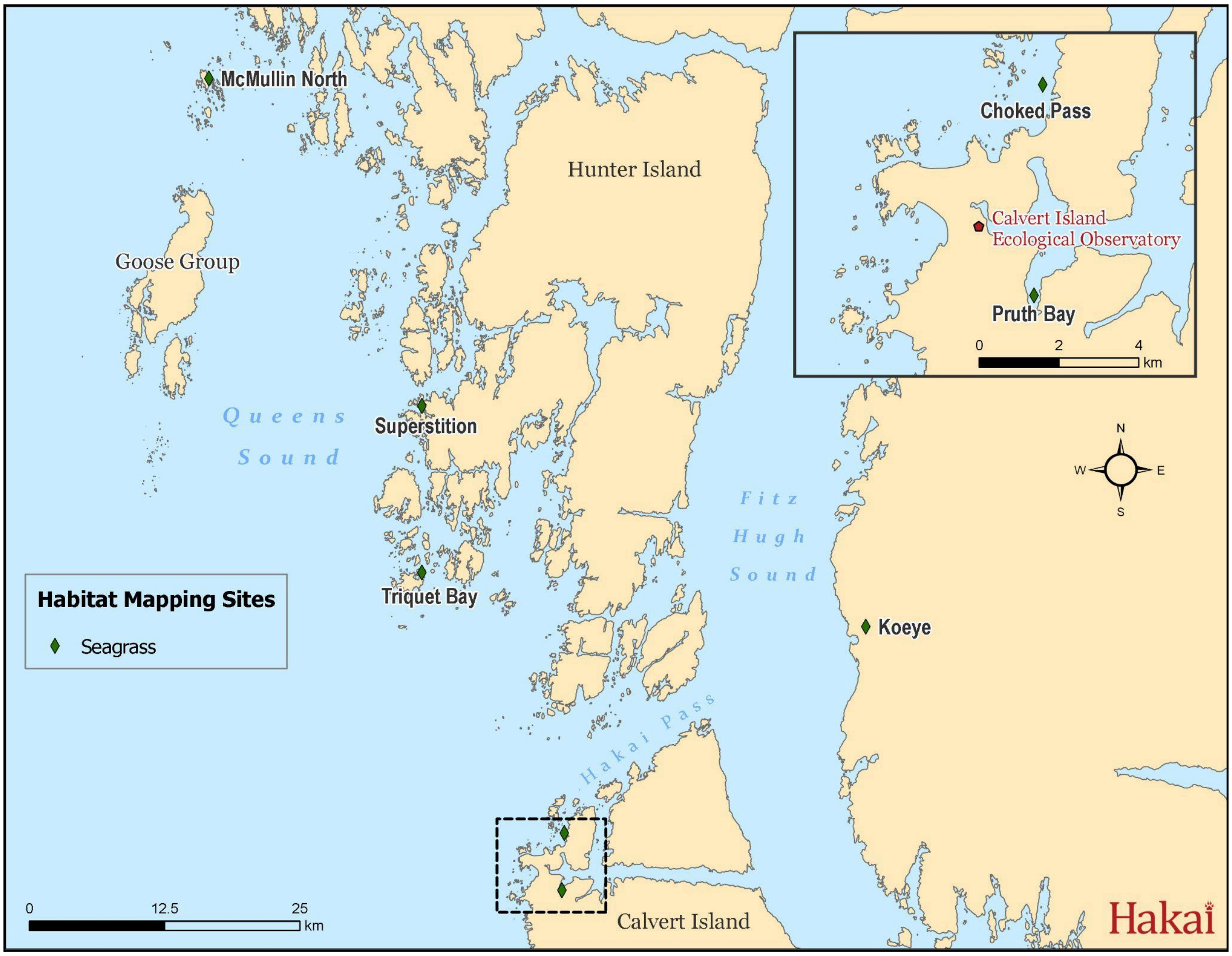

Figure 1. Location of annual eelgrass monitoring sites.

Table 1. Summary of temporal coverage of drone surveys at Hakai eelgrass monitoring sites on the Central Coast (see locations in Fig. 1).

Site name	Annual drone surveys
Pruth Bay	2015-2025
Choked Pass	2015-2025
Koeye*	2015, 2017-2025
Triquet Bay	2016, 2018-2025

McMullin North**	2015, 2017, 2018, 2020-2025
Superstition	2019-2025

*Incomplete drone coverage due to fog at Koeye in 2019 - AOI of coverage is provided in the data package.

***Incomplete drone coverage at McMullin North in 2022 - AOI of coverage is provided in the data package.

Methods

Imagery collection

Drone surveys are conducted annually at each long-term monitoring site on the Central Coast, typically from mid-July to early August in order to capture the peak extent of the eelgrass meadows. Drone flights are conducted at heights typically ranging from 95-120 m (above surface level) to ensure proper coverage of terrestrial features to generate accurate orthomosaics. These flight heights also ensure that flights are within the range allowed with a Basic Operations Pilot Certificate. In some cases, flights are conducted at higher altitudes under a Special Flight Operations Certificate. All flights are conducted at tide levels below 1 m (chart datum) in order to improve mapping of the subtidal edge.

Hakai Institute has and continues to use DJI drones for field surveys (DJI Phantom Pro 2, Pro 3, Pro 4, Pro 4V2 and Mavic 3E). Drone models for each survey are provided in the vector feature attribute table. Flights are conducted based on predetermined flight plans to ensure coverage of the entire study area, sufficient resolution as well as overlap between photos (minimum side and front lap of 70%).

All flights are logged with unique mission IDs (referred to as MOBE IDs) which allows the final data product to be linked to a unique flight.

Hakai Institute drone pilots operate under required Transport Canada requirements and have research permits to conduct surveys within BC Parks regions. Research plans and reports are reviewed on an annual basis with First Nation partners (Heilstuk Integrated Research Management Department and Wuikinuxkv First Nation).

Orthomosaic production

Orthomosaics of each site are created using a Structure from Motion Multi-View Stereo (SfM-MVS) workflow using Agisoft Metashape Pro running on Windows 11.

In the SfM-MVS workflow, common points (individual terrain features) are identified from multiple overlapping images (Figure 2). These “keypoints” allow different images to be matched and the scene geometry reconstructed (i.e. generates a 3D model). The images are projected onto the 3D model to remove the perspective distortion from the images. The final product is an orthomosaic. Orthomosaics are created more quickly using SfM-MVS than with traditional photogrammetry (Carrivick et al, 2016). The positioning data are obtained from the GNSS onboard the RPAS which have a horizontal positional accuracy of 5-10 m.

When a new site is established, an RTK survey is conducted with ground control targets. Final accuracy of the orthomosaics in this case is typically <0.25 m horizontal accuracy. All subsequent orthomosaics are georeferenced to the reference orthomosaic.

Final orthomosaics are in NAD83 CSRS UTM Zone 9 projection.

Eelgrass classification

The extent of eelgrass at each site was delineated using a combination of object-based image analysis (OBIA) (Blaschke, 2010) with eCognition Developer software (eCognition Developer 9, 2014) and manual delineation within ArcMap (version 10.6). Segmentation parameters used within eCognition were: scale 50, shape = 0.05, compactness = 0.5. The segmentation output from eCognition was imported into ArcGIS and manually classified as eelgrass. Where the segmentation performed poorly, the segments were removed and redrawn manually (common in deep water regions).

Ground-truth data from towed underwater video are used (where available) as an additional data source to aid in image interpretation.

The final shapefile has a 1 m smooth (PAEK) applied. Additionally, all eelgrass areas within 1 m of each of are aggregated (minimum patch area is 1 m^2, minimum hole size is 4 m^2).

Environmental characteristics during image acquisition (e.g. cloud cover, tidal height, wind speed) and site characteristics (e.g. eelgrass patchiness and density, presence of other non-eelgrass vegetation) have been shown to affect the confidence by which eelgrass is delineated from RPAS imagery (Nahirnick et al., 2018). We report mapping confidence level (high, medium, low) to describe overall eelgrass mapping reliability for each site.